



Enterprise Asset Management (EAMS) v15

Work Planning Module

End User Documentation

Update 12/30/2024

Work Planning Module Overview

The Work Planning Module is an optional and licensed business management method of identifying and evaluating activities that an organization performs, using activity-based costing to carry out a value analysis to improve strategic and operational decisions in an organization. It is also known as Activity Based Management. By evaluating planned versus actual work, organizations can optimize the planning and allocation of resources.

While a work order shows the work completed, work planning shows what needs to be accomplished with the resources available. Work Planning builds on what the work order captures from a higher perspective. The work plan seeks to analyze key activities that the organization performs and not just capture costs of their execution, but to establish goals for future accomplishment. These future goals for activities and established for accomplishment value and for expected cost, based on available resources. The Plan can then be compared to actual accomplishment and cost to determine if the organization is on track to meet the established goals, or if resources should be adjusted to gain efficiency.

Due to the complexity of initial setup and ongoing maintenance, it is strongly recommended that MaintStar provide training and system implementation assistance for this module.

Planning and Budgeting for Municipalities Overview

MaintStar's Work Planning System is specifically designed for municipalities to track varieties of daily activities, such as roads, sewers, traffic signs and other infrastructures repairs/replacements. This module is designed as a financial tool and is activities driven. It provides a great degree of detail for budgeting purposes, and in addition to planning capabilities, tracks actual amount of work performed against what was planned or budgeted.

The infrastructure planning capabilities includes specific equipment requirements, (trucks, bulldozers, sweepers, etc.) materials needed (concrete, paint, etc.) as well as crew size planning. It also tracks crew productivity and provides automatic cost roll up to specific city department, subdivision, multi-level cost center, area etc.

The planning capabilities provide a great degree of built-in flexibility and can be implemented on weekly, monthly, quarterly, or annual basis. It includes financial and budgeting components and should not be confused with a simple work order as it does more – it processes estimated work against actual.

Definitions

To project a budget for labor, materials and supporting equipment requirements, the following information needs to be entered and maintained in MaintStar:

- **Management Unit:** often a Department, Division, Project or specific cost center
- **Activities:** actual work or description of work to be accomplished, often tied to a unit of measurement, i.e. "HVAC Installation" with a measurement of "Each"
- **Work Quantity:** the total volume of work per specific type of jobs, which is called Activity. For example, in case of estimating road-paving activities, total miles of roads need to be paved
- **Quantity per unit:** unit of measure = 1 mile

- **Average Daily Productivity (ADP)** –how many units produced per day by crew. Example: 10 miles/day per crew
- **Crew:** group of people performing specific tasks
- **Total crew days required:** total work quantity/crew daily productivity
- **Unit Cost:** budget/total work quantity (including labor and materials)
- **Account Number:** Specific cost center-from MaintStar setup tables

Work Planning Example

In this example, there is a monthly plan to pave 100 miles of roads. The job to be performed by a maintenance crew consisting of:

- One supervisor
- Two paving specialists
- Two truck drivers
- Two maintenance workers
- 1 Dump truck = 3 hours a day
- 1 Paving vehicle = 8 hours a day

Budgeting period: The 'time' for which budget is setup and tracked (week, month, quarter, year)

Percentage of activities distribution: percentage of total labor, equipment required, and materials distribution per each activity for each Management Unit for specific budgeting period. For example, if there is a budget for 12 monthly periods, streets paving may represent 8% of total daily activities for an entire maintenance department. The maintenance department may perform other activities, such as streets sweeping, trimming trees, repairing road drains, etc.

- % default come from an activity code setup

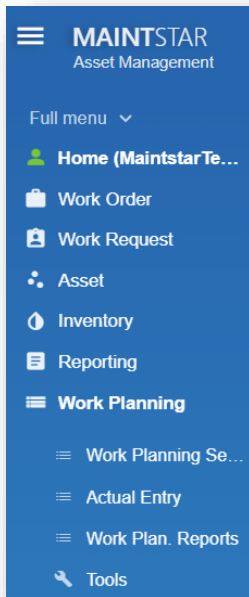
Each activity must have a specific unique I/D or code, for example

- Streets paving code: 0001
- Streets sweeping code: 0002, etc.

Labor days = total crew days x crew size / % per month. If there is a plan on a weekly basis, and a crew works 5 days a week, the total crew labor days = $5 \times \{ (2 \text{ people} \times 1 \text{ full day for 2 paving specialists} + 1 \times 1 \text{ paving vehicle driver}) + 2 \times 1 (2 \text{ helpers}) + 1 \times 0.25 \text{-dump truck driver which works only part time-one quarter of a day on this activities} \} = 5.25 \text{ man days} \times 5 \text{ days per week} = 26.25 \text{ crew total man days per week required to accomplish streets paving.}$

Work Planning Menus

The Work Planning Module is accessed from the Work Planning menu item in the sidebar, and contains the following (all of which will be covered throughout this document):

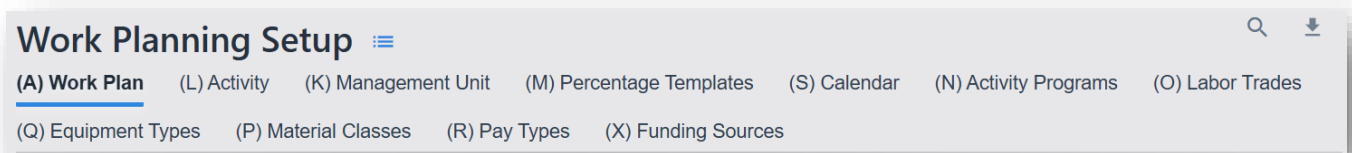


- **Work Planning Setup:** Contains the various setup and maintenance screens
- **Actual Entry:** Allows for direct entry of time NOT associated with Work Orders, i.e. meetings, training, time off, etc
- **Work Plan Reports:** Collection of pre-built reports specific to Work Planning. Reports can also be supplemented via Query and Reports
- **Tools:** contains calendar and plan rollover tools

Work Planning Setup Menu Overview

The Work Planning setup section has several tabs across the top of the screen. These tabs were separate menu items in earlier versions of MaintStar.

**The letter identifier is a reference from the previous version of MaintStar that many customers grew accustomed to and does not indicate any sort of precedence in setup.*



- **(A) Work Plan:** The primary screen for setting up the actual plans – this is used after setups have been completed.
- **(L) Activity:** Screen for creating and updating activities. This can also be accomplished via Setup Tables in Setup > Tables.
- **(K) Management Unit:** Screen for creating and updating Management Units
- **(M) Percentage Templates:** Setup for allocation of resources by percentage or days.
- **(S) Calendar:** Screen for setup of Fiscal (or Calendar) Years
- **(N) Activity Programs:** A collection of Activities (i.e. Roads Activity Program may contain all activities pertaining to Roads)
- **(O) Labor Trades:** Setup of Trades. This can also be accomplished via Setup Tables in Setup > Tables.

- **(Q) Equipment Types:** Allows for classification of equipment, for instance “Trucks”, “Heavy Equipment”, etc.
- **(P) Material Classes:** Allows for classification of materials, for instance “Paint”, “HVAC”, etc.
- **(R) Pay Types:** Setup of Pay Types. This can also be accomplished via Setup Tables in Setup > Tables.
- **(X) Funding Sources:** Allows for additional funding sources to be added

Work Planning Setup

The following is a suggestion for the initial setup of Work Planning. For the purposes of this documentation, there may be a slight variation in screens shown.

Calendar Setup

Start by setting up the fiscal year in the Calendar tab. This setup choice allows you to establish the budgeting time periods.

***Note:** If you check the check box to the left of a year line, a delete icon will appear next to the add button. **DO NOT DELETE** a line if it has valid data as it will not be retrievable.

Work Planning Setup 

(A) Work Plan (L) Activity (K) Management Unit (M) Percentage Templates **(S) Calendar** (N) Activity Programs (O) Labor Trades (Q) Equipment Types (P) Material Classes (R) Pay Type

 Add  

Budget Factors - 10% = 1.10, 25% = 1.25, 50% = 1.50

☐ Edit	Year ↑	Start Date	Period #	Freq	Unit	Default	Date Format	Manual Entry	Def. Hours	Labor %	Material %	Equip %	Lump Sum %	Non GFund %	Percentage Tmp
☐ Edit	2025	07/01/2024	12	1	Months	☒	mm/dd	Percent	8	1.00	1.00	1.00	1.00	1.00	Standard Template
☐ Edit	2024	07/01/2023	12	1	Months	☐	mm/dd	Percent	8	1.00	1.00	1.00	1.00	1.00	Standard Template
☐ Edit	2023	07/01/2022	12	1	Months	☐	mm/dd	Percent	8	1.00	1.00	1.00	1.00	1.00	Standard Template
☐ Edit	2022	07/01/2021	12	1	Months	☐	mm/dd	Percent	8	1.00	1.00	1.00	1.00	1.00	Standard Template

Current Year

8 hours

9 hours

10 hours

Year: 2025

Edit	No.	Work Days	From	To	8 hrs	9 hrs	10 hrs	Ttl Days	Weekend	Holidays
Edit	1	23	07/01/2024	07/31/2024	23	20	18	31	8	0
Edit	2	22	08/01/2024	08/31/2024	22	19	17	31	9	0
Edit	3	21	09/01/2024	09/30/2024	21	18	16	30	9	0
Edit	4	23	10/01/2024	10/31/2024	23	20	18	31	8	0
Edit	5	21	11/01/2024	11/30/2024	21	18	16	30	9	0
Edit	6	22	12/01/2024	12/31/2024	22	19	17	31	9	0
Edit	7	23	01/01/2025	01/31/2025	23	20	18	31	8	0
Edit	8	20	02/01/2025	02/28/2025	20	17	16	28	8	0
Edit	9	21	03/01/2025	03/31/2025	21	18	16	31	10	0
Edit	10	22	04/01/2025	04/30/2025	22	19	17	30	8	0
Edit	11	22	05/01/2025	05/31/2025	22	19	17	31	9	0
Edit	12	21	06/01/2025	06/30/2025	21	18	16	30	9	0

CHECK LIST

1. Setup NEW YEAR Calendar (with all default information and make it current).
2. Setup default working days, working hours.
3. Setup default Percentage Template and update the Calendar.

If you want to change DATE or FREQUENCY, DELETE the line then ADD a new one.

1. Click on Add button on a tool bar, then enter budget year. For example, 2021.
2. Enter starting day of the fiscal year. Example 01/01/21

3. Enter number of budgeting periods in the period # column, for example 12 for the monthly periods, or 52 for a weekly. NOTE: please make sure you decide to go with weekly or monthly before setting up and saving your work.
4. Enter 1 in the Frequency number column if we budget on a monthly or weekly basis. It means one period per month or per week. Note: if you budget on a quarterly basis you may enter 3 since there are three months in a single quarter.
5. Using drop down selection in the Unit column select budgeting units (weeks, months, etc.)
6. Click in the Default check box to indicate that this is a current budget we are going to work on.
7. Select from the date format drop down lookup the date format you want to be displayed
8. Manual Entry: there are two options here.
 - Crew Days: will be calculated automatically base on the current month and year when the Current Year button and the Hours button is clicked.
 - Percentage: workdays shown in the year details grid at the bottom of the screen, are calculated based on percentage template that is setup under tab 'Percentage Templates M.'
9. Leave the Budget Factors blank for now, it will be explained later
10. Default Template: you will need to setup the template under Tab M and update it here.
11. Click on Current Year button to select the working days for each month. You can change the working days by changing the numbers directly and working hours for each day by clicking on the hour buttons and view the changes. Click on Save Changes button if you want to keep the changes and click on Save button on the top of the current year table.
12. Number of monthly working days can be edited from this screen by clicking on specific month
13. In the Budget Factors columns enter any additional cost required if needed for this budget to be added on top for labor, equipment, material, non-general funds, etc. (often left as default)
 - For instance, if you need to increase 10% to the total labor cost, you will put 1.10. So, if your total labor cost is \$100.00, then the total labor cost for the budget is \$110.00.
14. Click Save when Complete
15. Repeat for past/future years if desired

System Equipment Setup

System Equipment Type Setup allows to setup equipment types or groups based on its functionality which will be used for planning. For example, using dump trucks, sweepers, and paving equipment for performing specific activities. This is the same setup choice as under Setup > Tables. Additional equipment types can be added if required. Enter hourly rate for equipment usage and estimated percentage of downtime (for example: a dump truck on average is used at 89% of its capacity, the rest 11% of time is spent on repairing it and performing a PM tasks.

Work Planning Setup											
(A) Work Plan (L) Activity (K) Management Unit (M) Percentage Templates (S) Calendar (N) Activity Programs (O) Labor Trades (Q) Equipment Types (P) Material Classes (R) Pay Types (X) Funding Sources											
☐ Add ☐ Edit ☐ Edit ☐ Edit	Asset Type	Workgroup	Description	Rate/hr	DownTime%	Billing/Hr	Remarks	Added By	Added Date	Updated By	Updated Date
	Tool	FACILITIES				0.00		Josh	09/16/2022	JGRIISSER	11/15/2022
	Vehicle	FACILITIES				0.00		CONVERT	01/09/2022	Josh	09/16/2022

1. Click Add to Add a new Type
2. Enter Billing Rates, and Remarks if required then Save upon completion.
 - Rate/Hr.: will be used under option A. (work planning)
 - Down Time %: will be used under option B which is in the report section “Recourse Requirement report’ for Equipment only.
 - Billing/Hr: will be used under the “Actual Entry,” menu section. This is the same as Billing Rate for Equip Type under Setup.

Materials Setup

This menu choice allows to setup categories of materials to be used. For example, for road repairs different type of concrete may be used. Here we will identify just a generic group of materials - called Class.

Work Planning Setup

(A) Work Plan (L) Activity (K) Management Unit (M) Percentage Templates (S) Calendar (N) Activity Programs (O) Labor Trades (Q) Equipment Types **(P) Material Classes**

☐ Add	Class	Description	Unit Name	Unit Cost	Remarks	Added By	Date Added	Updated By	Date Updated
☐ Edit	Construction					Josh	09/19/2022		
☐ Edit	Decorating					Josh	09/19/2022		
☐ Edit	Electrical					Josh	09/19/2022		
☐ Edit	Expendables					Josh	09/19/2022		
☐ Edit	Hardware					Josh	09/19/2022		
☐ Edit	HVAC					Josh	09/19/2022		
☐ Edit	Janitorial					JGRIISSER	10/27/2022	JGRIISSER	10/27/2022
☐ Edit	Plumbing					Josh	09/19/2022		
☐ Edit	Safety					Josh	09/19/2022		

1. Click on New, then enter material class, description, then unit name (example concrete we would measure in cubic feet, then cost per unit.
2. Click Save

Labor Setup

This table is the setup for Trades and Trade rates. Trades can also be set up from the Setup > Tables. For example, Lead Utilities Operator \$96.49/hr. Click on New, then enter labor trades and rates. Click on Save when finished.

Work Planning Setup

(A) Work Plan (L) Activity (K) Management Unit (M) Percentage Templates (S) Calendar (N) Activity Programs (O) Labor Trades (Q) Equipment Types (P) Material Classes

Add								
☐ Edit	Trade	Description	Rate/Hr	Remarks	Added By	Date Added	Updated By	Date Updated
☐ Edit	ACC	Accounting Clerk			JGRIISSER	09/30/2022	JGRIISSER	09/30/2022
☐ Edit	ACS	Accounting Specialist			JGRIISSER	09/30/2022	JGRIISSER	09/30/2022
☐ Edit	BMS	Building Maint. Supervisor			JGRIISSER	09/30/2022	JGRIISSER	09/30/2022
☐ Edit	CLC	Contract/Leasing Coordinator			JGRIISSER	09/30/2022	JGRIISSER	09/30/2022
☐ Edit	CPC	Contract/Project Coordinator			JGRIISSER	09/30/2022	JGRIISSER	09/30/2022
☐ Edit	CTW	Certified Tradesworker			JGRIISSER	09/30/2022	JGRIISSER	09/30/2022
☐ Edit	DBC	Database Coordinator			JGRIISSER	09/30/2022	JGRIISSER	09/30/2022
☐ Edit	DIVMGR	Division Manager			JGRIISSER	09/30/2022	JGRIISSER	09/30/2022
☐ Edit	FIM	Financial Manager			JGRIISSER	09/30/2022	JGRIISSER	09/30/2022
☐ Edit	FMT	Facilities Maintenance Tech			JGRIISSER	09/30/2022	JGRIISSER	09/30/2022
☐ Edit	PGM1	Program Manager I			JGRIISSER	09/30/2022	JGRIISSER	09/30/2022
☐ Edit	PGM2	Program Manager II			JGRIISSER	09/30/2022	JGRIISSER	09/30/2022
☐ Edit	PRM2	Project Manager II			JGRIISSER	09/30/2022	JGRIISSER	09/30/2022
☐ Edit	STA	Staff Assistant			JGRIISSER	09/30/2022	JGRIISSER	09/30/2022
☐ Edit	TL	Team Leader			JGRIISSER	09/30/2022	JGRIISSER	09/30/2022
☐ Edit	TW	Tradesworker			JGRIISSER	09/30/2022	JGRIISSER	09/30/2022

Activity Programs Setup

(Planning menu, Option N.)

This screen must be done here in Work Planning module.

MAINTSTAR

Asset Management

Full menu

Home (BSMITH)

Work Order

Work Request

Asset

Asset Management

Inventory

Reporting

Work Planning

Work Planning Setup

Actual Entry

Work Plan. Reports

Tools

Purchase Orders

Invoicing

Fleet

Setup

Security

WORK PLANNING SETUP

(A) Work Plan

(L) Activity

(K) Management Unit

(M) Percentage Templates

(S) Calendar

(N) Activity Programs

(O) Labor Trades

(Q) Equipment Types

(P) Material Classes

(R) Pay Types

(X) Funding Sources

ADD

☐ Edit	Program Code	Description	Added By	Date Added	Updated By	Date Updated
☐ Edit	1	ADMINISTRATION	Convert	06/14/2019		
☐ Edit	10	LANDSCAPE MAINTENANCE	Convert	06/14/2019		
☐ Edit	11	IRRIGATION MAINTENANCE	Convert	06/14/2019		
☐ Edit	12	WEED ABATEMENT	Convert	06/14/2019		
☐ Edit	13	RECREATION FIELD MAINTENANCE	Convert	06/14/2019		
☐ Edit	14	PARK MISCELLANEOUS	Convert	06/14/2019		
☐ Edit	15	TREE MAINTENANCE	Convert	06/14/2019		
☐ Edit	16	BUILDING MAINTENANCE	Convert	06/14/2019		
☐ Edit	17	CUSTODIAL SERVICES	Convert	06/14/2019		
☐ Edit	18	POOL MAINTENANCE	Convert	06/14/2019		
☐ Edit	19	MISCELLANEOUS	Convert	06/14/2019		
☐ Edit	2	ASPHALT MAINTENANCE	Convert	06/14/2019		
☐ Edit	20	MISC UTILITY MAINTENANCE	Convert	06/14/2019		
☐ Edit	21	MISC LIGHTING	Convert	06/14/2019		
☐ Edit	3	SIGNING AND STRIPING	Convert	06/14/2019		
☐ Edit	4	ROAD MISCELLANEOUS	Convert	06/14/2019		
☐ Edit	5	CONCRETE MAINTENANCE	Convert	06/14/2019		
☐ Edit	7	WATER	Convert	06/14/2019		
☐ Edit	72	WATER MISC	Convert	06/14/2019		
☐ Edit	8	SEWER MTCE	Convert	06/14/2019		

A group of individual activities is called a program. Example, street repair can be a Program. It consists of a variety of activities (tasks), such as sweeping, paving, removing old pavements, etc.

- Deleting a line: highlight a line and click on delete (scissors). If the line is blank, save it then delete.

Activity Percentage Distribution Template (Planning menu, Option M.)

No	Period/Date Range	Season	Percentage
1	01/01-01/31	Winter	18.23
2	02/01-02/28	Winter	7
3	03/01-03/31	Spring	10
4	04/01-04/30	Spring	7
5	05/01-05/31	Spring	6
6	06/01-06/30	Summer	8
7	07/01-07/31	Summer	8.00
8	08/01-08/31	Summer	8.00
9	09/01-09/30	Fall	8
10	10/01-10/31	Fall	8.00
11	11/01-11/30	Fall	3.5
12	12/01-12/31	Winter	8.27
Total %			100.00

This template will be used under Option S (calendar setup) and Option K. Basically, this menu choice allows to replace a tedious repeatable data entry by setting up a template which can be reused. This template is used to set up the percentage of activities for each budgeting period. For example, streets sweeping represents 8% of total monthly work. You may establish different percentages for different periods for example June sweeping represents 8% of total work, but in July due to upcoming holidays and big crowds it will represent 12 % of total work.

You may set up a template for sweeping with 8% activities per each month must be total to 100% for the entire year. Enter template id in a template field, then enter % per each month, then click on Save button on the right side of the window to save the template. You can save multiple different templates as you need. Templates are yearly dependent. When you are in this screen, the options you have are the buttons on the right-hand side.

Note: you can enter your own numbers if it adds up totally 100. The number is the percentage number

Setup Buttons and Options

- New/Add: create a new template
- Calculate Highlight Rows: highlight (Ctrl + left mouse clicks) the months you want to calculate percentage of work equal to 100%, then click this bottom.
- Save: to save your template and changes.
- Select/Return allows you to select a different template then apply to the plans by clicking on Get Default icon from another screen such as Option L.

- Whole Number: if this box is checked, you can only enter an integer in the percentage fields. No decimal numbers.

Pay Type

The Pay Type table allows you to create different calculations of hourly cost for Labor pay. When labor entries are made for a technician in a work order or in direct entry, you can use the pay type field to select a pay type from this table to be applied to that entry. The formula you see here will adjust the labor rate for the labor entry.

1. Click add to add a new pay type.
2. Give it a name and description.
3. You have three ways to manipulate the standard base pay rate:
4. HrFactor Pay = This value is a multiplier to the base pay rate. 1.0 is no change, 1.5 is time and a half, 2.0 is doubletime.
5. Hrfactor Xtra= This value will add to hourly calculation. Example Susan has an hourly pay rate of \$25.00. If the pay rate we apply has a value of 2 in the Hrfactor Xtra, Susan's rate will be calculated at \$27.00 per hour (\$25+2).
6. Flat rate= This adds a flat amount for each entry. It would be used if you have a call out bonus paid if a technician is called out during the middle of the night etc. A Flat rate of \$200.00 will add \$200.00 to the normal hourly rate for a labor entry. Example John was called out during the night to solve a crisis. He worked for 4 hours. His hourly rate is \$30.00. $30 \times 4 = 120 + 200$ flat call out rate = \$320.00
7. Put a check mark in your default pay rate type checkbox, and it will be applied to labor entries by default.

The screenshot shows the MAINTSTAR Asset Management interface, specifically the 'WORK PLANNING SETUP' section. The 'Pay Types' tab is active, displaying a table with the following data:

Pay Types	Pay Description	HrFactor Pay	HrFactor Xtra	Flat Rate	Hr Info	Remarks	Default Type	Added By	Date Added	Updated By	Date Updated
Call Out	Call out rate	1.00	0.00	200.00			☐	BSMITH	02/01/2021		
LEAVE	LEAVE TIME	1.00	0.00	0.00	0		☐	Convert	06/14/2019	Admin	07/21/2020
OTP	OVERTIME PAY	1.50	4.00	1.00	1		☐	Convert	06/14/2019	BRETT	08/25/2020
Reg	Reg Pay	1.00	0.00	0.00			☒	ERIC	07/09/2019	BRETT	08/25/2020

Activity Description (Planning menu, Option L.)

This menu choice allows you to create Activity Codes. An Activity code describe in plain English specific activities steps. The Activity code is a key element in Work Planning. It tracks “what work was performed”. The Activity code can be linked to the Activity Programs. Multiple Activity Codes can be link to one Activity Program.

For instance, we have Activity Program of Road Repair. We can link different Activity Codes to this Activity Program such as Sweeping, Patching, Curb Painting, etc.

Note: always save before leaving the screen.

To change the Activity Distribution Percentage:

1. Click on Get default icon on the tool bar and select distribution template (from the previous (M) menu choice) to link to this activities by using Select/Return button, then click OK to save.
2. If you made any changes to the Activity after creating the Work Planning, you must select other activity, save; then re-select the activity, and Save to apply the new changes.
3. If you made changes to the description after creating the Working Planning, it will not apply the current work planning. You can add the descriptions directly on the work planning by clicking on the Edit Task, save it, then click on Edit Task again to close. Another way to apply the text changes is to use option U.

The screenshot shows the MAINTSTAR WORK PLANNING SETUP interface. The left sidebar contains a navigation menu with options like Asset Management, Inventory, Reporting, Work Planning, and Tools. The main area is titled 'WORK PLANNING SETUP' and includes tabs for (A) Work Plan, (L) Activity, (G) Management Unit, (M) Percentage Templates, (S) Calendar, (N) Activity Programs, (O) Labor Trades, (Q) Equipment Types, (P) Material Classes, and (R) Pay Types. The 'Activity' tab is selected, showing fields for Activity Code (205), Description (CRACK SEALING), Activity Program (2 - ASPHALT MAINTENANCE), Funding Source (N/A), Work Location, Inventory Unit (LN MILES - LN MILES), Work Unit (POUNDS - POUNDS), and Work Description. A 'Planning Criteria' section is also present. At the bottom, there is an 'Activity Distribution Percentage' table with columns for Total %, 01/01, 02/01, 03/01, 04/01, 05/01, 06/01, 07/01, 08/01, 09/01, 10/01, 11/01, 12/01, and a final column for 01/01. The table shows values for each month, with a total of 111.23. Annotations with red callouts point to specific features: 'Open Search' points to the search icon, 'Move through search result list' points to the search results list, and 'Drag Corner to expand window' points to the corner of the 'Work Method' panel.

Open Search

Move through search result list

Drag Corner to expand window

Total %	01/01	02/01	03/01	04/01	05/01	06/01	07/01	08/01	09/01	10/01	11/01	12/01
111.23	23.23	8.00	9.00	8.00	8.00	9.00	8.00	8.00	9.00	8.00	4.00	9.00

- Activity Code: example 0001 for paving, then description
- Activity Program: (was previously established, example roads repairs), M option.
- Inventory Unit: this was done in MaintStar Setup Module, Unit of Measure (example linear foot). The Inventory Unit can be typed in on this screen only. Inventory Unit and Work Unit don't necessarily have to be the same.
- Work Unit: this was done in MaintStar Setup Module, Unit of Measure
- Same as Labor: if this box is checked, the qty work unit will be calculated the same as the unit of the labor cost. Normally, in most cases this box should not be checked; but, for Meetings example above, this box should be checked since you don't have the quantity for meetings; so, the qty is the 'same as labor'
- Work Location: comes from Maintstar Setup Module, locations setup
- Low and High: not used at this point

The four windows below are text instructions for the project. See screen shot above. You must fill them out before going to Option A, or you can leave it blank and fill it in later under Option A., Edit Task.

- Description of Work
- Planning Criteria
- Work Method, Check Point
- Note

Management Unit (Previously called -Activity System) Management Unit Setup
(Planning menu, Option K.)

Note: each level MUST be saved before next level is entered. Each level has its own Labor, Equipment, and Material Available.

The screenshot shows the MAINTSTAR WORK PLANNING SETUP interface. The left sidebar contains a navigation menu with options: Full menu, Home (BSMITH), Work Order, Work Request, Asset, Asset Management, Inventory, Reporting, Work Planning (selected), Work Planning Setup, Actual Entry, Work Plan, Reports, Tools, Purchase Orders, Invoicing, Fleet, Setup, and Security. The main content area is titled 'WORK PLANNING SETUP' and features a series of tabs: (A) Work Plan, (L) Activity, (K) Management Unit (selected), (M) Percentage Templates, (S) Calendar, (N) Activity Programs, (O) Labor Trades, (Q) Equipment Types, (P) Material Classes, and (R) Pay Types. Below the tabs, there is a search bar and buttons for '+ NEW', '+ SAVE', and '+ DELETE'. The 'Management Unit' section displays a tree structure with levels 1 through 8. Level 1 is 'Public Works', Level 2 is 'Streets', and Level 3 is 'Concrete Maintenance'. To the right of the tree, there are input fields for 'Labor Available', 'Equipment Available', and 'Material Available', each with an 'ADD' button. At the bottom, there is a 'Percentage' table with columns for Weekends, Holidays, Working Hrs per Day, and a grid of numerical values.

	Weekends	Holidays	Working Hrs per Day														
Edit	☐	☐		21.00	20.00	23.00	22.00	21.00	22.00	22.00	22.00	22.00	21.00	22.00	23.00		

This portion of setup may be quite challenging to comprehend. We recommend on-site assistance and training to provide proper guidance as well as insurance that it is done right.

The Management Unit is a way to structure a numerical code sequence which represents hierarchical levels in the structure of the organization. The code structure indicates the level and relationship of each work group in the hierarchy. The numerical coding has a purpose and must be created in the following format.

When we do work planning, MaintStar allows us to plan for multiple departments which may be structured in hierarchical order. Example the planning to be done for a department (first level) which consists of the following sub departments or units (level 2). For instance, under "Public Works" you may find:

- 1-Streets
- 2-Parks
- 3-Sewer
- 4-Waste Water and so on...

The streets department may consist of two following sub departments (level 3)
Highways, Local Roads, and so on...

We can budget for the entire Public Works with drill down up to 8 levels. With such capabilities you may budget for practically any size of operation (remember, to locate any individual in the entire 6 billion universe we may need just about 7 levels-Country, City, Street, etc.)

Management Unit (MUST BE UNIQUE). It can be a department of organization for which we are setting the plan. Management Unit can be also a specific project, which needs to be tracked. It is identified by the Management Number which is a specific account or cost center where cost is charged. It can be broken down in hierarchical order up to 8 levels-for management reporting. Make sure to click on Save (OK) button for each administrative Number or before exiting the screen.

The Management Unit Number (accounting number) consists of 8 levels (displayed at a top of the screen-8 fields). We use dashes in this example to show different levels of reporting structure and indicate separate fields of entry.

Each layer down in the organization adds an additional digit to the Management Unit number. The example of a Management Unit Number is 111PB can be entered in the first left field (up to 5 alpha-numeric characters)-we will call this field first level. For example, 111PW can be Public Works Department. Then one alpha numeric character can be entered in each of next 7 fields. -we will call these sub levels

If we add another character (1) to the next field, we will have for example:

Streets sub department UNDER Public Works	111PW-1
Under Streets sub department-Signs department	111PW-1-1
Under Streets sub department-General Maintenance sub department	111PW-1-2
Parks sub department UNDER Public Works	111PW-2 and so on.

Example: We entered 111PW-1, then Save.

To enter the next Management Unit #, 111PW-2 just override the last character of previous entry, for example character 1 with 2, then click on Save. Now we have entered two accounting numbers 111PW-1 and 11PW-2.

To view the previous Management Unit number, use Green left pointing arrow. To view the next Admin No. use right Green arrow or click on the Open Search to search for other Management Units.

Each level of Management Unit Number (Cost Account No) needs to be linked with the corresponding Management Unit (project, for example).

Basically, Management Unit as mentioned above is a project, which needs to be tracked. The cost will be charged to specific Accounting Number – we call it Management Unit Number which we just finished setting up in our example. Now we are going to set up a linkage between Management Unit and corresponding Admin. Number. Another words, we need to indicate which projects are charged to which accounts.

Please enter all Assets in the asset section which are identified as equipment. They are created in the asset section of MaintStar. Equipment use will be planned for and tracked in the Work Plan. Equipment would be vehicles such as trucks, graders, dozers, and other heavy equipment. Equipment could also be smaller items such as mowers, generators etc.

Then come back and using Drop down windows select specific Equipment-(right side of the screen) for displayed Management Unit Number Streets Repairs. Then save.

To assign more Systems to next Management Unit Number, override information in a Management Unit Number field by typing it over, then link is with Management Unit using drop down window, then Save after each entry. This linkage is mandatory.

Note: Each System linked to specific accounting level can be entered only once and cannot be reused for a different Administrative Number. All linkages must be unique. Example Project A (Management Unit A) will be always charged to Administrative Number 11PW-1. You cannot charge the same project A to another Admin. Number 111PW-12, or 11AB-124,etc.

Deleting or Changing Management Unit

Go to Management Unit menu, open the search link, select the Management Unit you need to delete or change, click the check box on the line for the management unit you want to delete. The delete icon will appear over the list. click the delete trash can to delete.

After entering Management Unit information, we can proceed to the next step. At the bottom of the screen, we show 12-month calendar (can be also setup on a weekly basis-up to 52 weeks). This information comes from Calendar setup (Planning menu, sub Option R). This calendar can be set up based on 8 working hours per day-default. Or 9 and 10 hours per day if required. This calendar also indicates number of working days per month (excluding weekends)

If you check boxes on the left side, it will include weekends and holidays. The holidays schedule setup is under Maintstar Security module). You may override the number of working days available per each month right on the screen, then Save.

Entering Labor, Equipment, and Material Resources.

The next step is to enter departmental resources required to each level to perform all jobs. Basically, we will enter labor available (Labor Trades and number of people per trade, then Materials available and Equipment available to perform a job. For example, Streets department has 10 drivers, 3 electricians, 5 sign inspectors and 10 trucks.

For each accounting level you can set up this information. The setup for each level or department is independent, so we will not roll up labor and equipment resources at this moment to an upper level. It will be done through reporting later.

Click on specific administrative accounting field (up to 8 levels) then click on Tab -Labor Available-top of the screen. The labor entry screen will be displayed. Click on New, then enter specific trades and quantity of people per trade. Upon completion click on Save. You may enter as many trades as you like. Upon completion Click on Equipment Available Tab and enter equipment information. Upon completion Click on Materials Available Tab and enter materials information. Repeat this entry for each accounting Administrative Number code AND each Administrative System. Exit the current window or the previously open window.

The first part of the general setup is completed. Now we are ready for a Planning Screen Setup

Work Planning Screen Setup

Planning Menu, Option A.

- The bottom section shows the budgets for the whole Management Unit that includes one or more activities, with the total Labor Days for all Activities, and total Daily Employees for all activities. The Planned Budget in the middle right hand side is the planned budget for one activity of the current screen.
- The middle section shows the percentage distribution template, labor days, and crew days for the current activity.
- The Left and Right Green Arrows will display all activities within the Admin System, or the next admin system.
- Edit Task icon: allows you to edit the activity tasks. Click on the icon to edit. Click on the icon again to turn it off.

Type: select Activity

- Code: you must type in the exact activity code that was done under option K
- Image Location: double click on the line and look for your image, then Save and exit

WORK PLANNING SETUP

(A) Work Plan (L) Activity (K) Management Unit (M) Percentage Templates (S) Calendar (N) Activity Programs (O) Labor Trades (C) Equipment Types (P) Material Classes (R) Pay Types

(X) Funding Sources

Open Search

+ NEW + SAVE + DUPLICATE + DELETE

Work Plan No. 0000001189 2021 01/26/2021

Mgmt Unit 00003 - Public Works

Activity code 100 - OUTSIDE CONTRACTOR

Activity Program null - null

Labor / Equipment / Material

Edit	No	Type	Code	Description	Qty	Rate	Cost	UOM
☐	1	Labor	AA	ADMINISTRATIVE ASSISTANT	2.00	44.44	88.88	HR
☐	2	Equip	Car	4 door sedan 123	10.00	20.00	200.00	HR
☐	3	Material	1/2AC	1/2 ASPHALT CONC	10.00	50.00	500.00	TON

Labor: \$88.88, Equipment: \$200.00, Material: \$500.00

GET DEFAULT

	Total %	01/01 01/31	02/01 02/28	03/01 03/31	04/01 04/30	05/01 05/31	06/01 06/30	07/01 07/31	08/01 08/31	09/01 09/30	10/01 10/31	11/01 11/30	12/01 12/31
Percent	99.96	8.33	8.33	8.33	8.33	8.33	8.33	8.33	8.33	8.33	8.33	8.33	
Labor Days	801.00	66.75	66.75	66.75	66.75	66.75	66.75	66.75	66.75	66.75	66.75	66.75	
Crew Days	400.50	33.38	33.38	33.38	33.38	33.38	33.38	33.38	33.38	33.38	33.38	33.38	

Mgmt Unit Budgets: \$1,125,821.52

	Total %	01/01 01/31	02/01 02/28	03/01 03/31	04/01 04/30	05/01 05/31	06/01 06/30	07/01 07/31	08/01 08/31	09/01 09/30	10/01 10/31	11/01 11/30	12/01 12/31
Labor	800.64	66.72	66.72	66.72	66.72	66.72	66.72	66.72	66.72	66.72	66.72	66.72	
Employees	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	

Work Units 100.00

Qty per Unit 4.0050

Total Work Qty 400.50

Daily Work Qty 1.00

Budget Factor

Unit Cost \$2,811.04

Time(Hr) per unit 16.00

Crew Size 2.00

Crew Days 400.50

Lump Sum Budget .00

Non General Fund .00

Planned Budget \$1,125,821.52

Click on the New button to obtain the Working Planning #.

Note: make sure to Select Admin Sys and Activity Code before clicking on Save button for the plan to be calculated properly.

Note: the system is designed as a planning financial and management tool, not as a fixed program.

Meaning that you can use this menu option to manipulate the program in different ways. Upon setup completion you would need to manipulate this information to obtain a specific result.

This screen provides an annual plan visual information on a monthly –12 months (or weekly-52 weeks) basis for labor days and cost distribution per specific activity.

The initial selection whether the planning is to be done on the weekly or monthly basis is under Planning Module Menu, Option R-Calendar setup. Please do not switch from weekly to monthly planning at your discretion or vice versa. This is a one-time setup. Decide up front and stay with it until the end of the planning year.

This screen allows us to link activity to each Administrative System, (specific department or project) so we can calculate later a monthly basis labor requirements per Administrative System.

An Management Unit may have multiple activities attached to it. . For example, for Streets departments the following activities are to be performed:

- Sweeping
- Streets repairs
- Signs and traffic lights inspection

Linking Management Unit with corresponding Activity

Let Maintstar know which activities need to be performed per Administrative System, a linkage needs to be established. You MUST complete all fields in Activity Code (option K) before linking to Management Unit. Click on New, select Management Unit using drop down window, then select Activity code. For instance, Management Unit = Streets, Activity Code=Asphalt Repairs, Sweeping, etc.

Labor Tab – Equipment Tab – Material Tab

Note: make sure to click on Save (OK button) for the Work Planning to be calculated properly. If changes are made elsewhere after the Work Planning has been done, make sure to click on Save for the work planning to be re-calculated.

Labor Tab (labor cost)

This is hourly rate, and Qty is 1 person, not 1 hour. Click on labor tab, then click on Add Detail icon on the top toolbar (icon looks like a clipboard) if you need to add a worker. Click on Trade drop down box to select a worker; description and rates should be filled in automatically, this was set up under option N. (Labor Trades). You can manually change the rate if you wish. To add another Trade just hit enter or click on Add Detail. Total Labor Rate will be multiplied by the hours a day based on the calendar setup template. $\text{Total Labor Cost} = \text{Total Rate} * \text{Hrs a Day} * \text{Crew Days}$.

To add additional workers, click on the Add Detail icon again. To delete worker, click on the worker, then click on the Del Detail icon (red scissors). Clicking on the wrong scissors will delete the Work Planning#, but you will be confirmed.

Equipment Tab (equipment cost)

This is hourly rate, and Qty is 1 equipment. Click on Equipment Tab, then click on Add Detail to add Equipment. Add and delete should be the same as Labor above. These pieces of equipment were set up under option P. To delete equipment, click on the equipment, then click on the Red Scissors. Total Equipment Rate will be multiplied by the hours a day based on the calendar setup template. $\text{Total Equipment Cost} = \text{Total Equipment Rate} * \text{Hrs. a Day} * \text{Crew Days}$.

Material Tab (material cost)

This is the daily rate. Click on Material Tab, then click on Add Detail to add material. These materials were set up under option O. Add and delete should be the same as Labor above. To delete a material, click on a material, then click on the Red Scissors. Material total rate will be multiplied by the crew days. $\text{Total Material Cost} = \text{Total Material Rate} * \text{Crew Days}$

Work Units

This is a yearly work units. Example: 100 trees a year, can be referred as inventory.

Qty per Unit

This is known as 'level of effort'. If it is 1, then you can trim 100 trees a year. But if you can only do 50 trees a year, then the qty per unit is 0.5. This is just one example. Different methods can be used depending on what and how you want to plan.

Total Work Qty

Work Units times Qty per Units. $(100 * 1) = 100$

Daily Work Qty (meaning a day of work qty, and it does not have to be everyday)

This is your estimation of how many trees you can do a day. Some call it ADP (Average Daily Productivity)

Unit Cost

Unit cost = planned budget / total work qty.

Unit Hours:

(next to Unit Cost) how long it takes to do one unit is:

$[(\text{Labor Days}) (8 \text{ hrs. a day})] / \text{Total Work Qty}$

Crew Size

Each worker is counted as 1, under Labor Tab, if you have 2 workers, then crew size is 2, or if you have 1.5 workers, then crew size is 1.5.

Crew Days

Equals Total Work Qty divides Daily Work Qty.

$\text{Crew Days} = \text{Total Work Qty} / \text{Daily Work Qty}$

These crew days then will be divided into 12 months. This is a yearly plan

Budget Factor: check this box if you want to apply your Budget factor that was setup under Option S.- Calendar Setup (scroll to the right)

Labor Days: equals crew size times crew days (labor days = crew size * crew days)

Employees = Labor Days of Budgets Month / Workdays of that month

This is the total number of all employees required for this period, for all activities, and for this admin system only.

Lump Sum Budget, Non-General Fund: enter these amounts into the boxes if you have any to be added to Planned Budget.

Budget Factor Box: if this box is checked, the total Planned Budget will be calculated based on your settings under Option R, and is calculated as follow:

- Labor (Budget Factor) = $(\text{Total Labor Cost} * \text{Hrs. a Day} * \text{Crew Days} * \% \text{Labor Factor})$, then add to the total Planned Budget (option S)
- Equipment (Budget Factor) = $(\text{Total Equip. Cost} * \text{Hrs. a Day} * \text{Crew Days} * \% \text{Equip. Factor})$, then add to the total planned budget (option S)
- Material (Budget Factor) = $(\text{Total Material Cost} * \text{Crew Days} * \% \text{Material Factor})$, then add to the total planned budget (option S)
- Lump Sum Budget (Budget Factor) = $(\text{Lump Sum Budget} * \% \text{Lump Sum Budget Factor})$, then add to the total planned Budget
- Non General Fund (Budget Factor) = $(\text{Non General Fund} * \% \text{Non-General Fund Factor})$, then add to the total planned Budget
- Planned Budget: this is the total amount of the budget for the whole year.

- $\text{Planned Budget} = [((\text{labor rate} * \text{Wk Hr per Day}) + (\text{equipment rate} * \text{Wk Hr per Day}) + (\text{material cost per Day})) * (\text{crew days})] + (\text{Lump Sum if any})$
- Add if required any overhead Lump Sum Cost and Non-General Funds Cost.

Enter work units and quantity per unit (example: total streets =100 each street is 10 miles Enter average daily productivity example 5 miles/day. The information can be entered only in yellow fields. Information in gray fields is for viewing only.

- Click on Drop down window and select trade. Enter hours required per activities. The rate will be displayed from the previous setup. Click on Add Details button on a tool bar to add another record if required.
- Click on Equipment Tab and setup Equipment required per attached activities. Click on Add Details button to add another equipment
- Click on Materials Tab and add materials required. Click on Save upon completion.
- To add more activities to the same administrative system, click on New button on a Tool Bar, then select the same Management Unit and then select next activities from a drop-down window. Then repeat above steps. Click on Save upon each entry completion.
- To view next activity for the same Management Unit use red arrows on a tool bar. If that was the last activity, you will be prompted for the next Management Unit displayed.
- To add the Tasks for the Work Planning, click on the Edit Task icon on the toolbar. Finish your tasks, then click on the Edit Task icon again to close. This task will be printed when you print out the plan. Printouts of the plans will show all your resource requirements used in the plan.
- To add new Management Unit, Click on New, then select Administrative System, then repeat steps above.
- The system automatically assigns a unique consecutive work plan number to each work plan sheet-top right. It also stamps date and time
- At the bottom the screen there are two windows are displayed.
- The following calculated information will be displayed in a gray fields (no editing)
- Crew Size, Crew Days, Unit Cost, Budget
- The middle window: 12 months or 52 weeks- shows planning information for a single activity
- The table is displayed with monthly (or weekly) information and shows projected percentage of a single activity scheduled monthly. For example, if we want to distribute activity evenly through entire year, we need to input 8,34% per months, which will total approx. 100% for an entire year.
- Labor days- shows total annual and monthly or weekly labor days required per single activity.
- The bottom window:12 months or 52 weeks- shows planning summary budget \$ information for all activities for the whole Management Unit
- Employees-Number of employees required per month or week to accomplish all activities. This number could have a decimal point example, 10,5 employees

- The setup is completed. The next step is entering actual daily data in a system to track completed work.
- Note: After the Work Planning is set up properly, the main three factor keys here are now the Work Units, Qty Per Unit, and Daily Work Qty. You will manipulate these three numbers to fit your plan. Let's look at a few examples: 100 meetings, 5hrs each meetings, 500hrs total. There are three different ways to manipulate these three numbers.

Example 1:

Total Hours:	500hrs	500hrs	500hrs
Work Unit:	1year	100meetings	5hrs
Qty Per Unit (effort):	500hr	5hrs a meeting	100meetings

As you can see total hours are 500hrs, but there are three different ways to manipulate these numbers.

A) Adjust the total hours for the whole year B) You want 100 meetings a year, but you want to adjust how many hours a meeting C)You want 5hrs a meeting, and change the number of meetings.

Example 2:

- Wk Units: 12 meetings
- Qty per Unit: 1 (hour per meeting)
- Total Wk Qty: 12 hrs of meetings
- Daily Work Qty: 16 hrs (assuming you have 2 workers, 8hrs a day. This is the maximum productivity of work they can do a day)
- Unit Hour: 1 (this has to be 1 when dealing with hours)

Example 3:

- Wk Units: 12 meetings
- Qty per Unit: 1 (1 day a month)
- Total Wk Qty: 12 days of meetings
- Daily Wk Qty: 2 days (assuming you have 2 workers, 8hrs a day. This is the maximum productivity of work they can do a day. Since unit is in days, 16hrs would be 2 days)
- Unit Hour: 1 (this has to be 1 when dealing with hours)

Example 4:

- Wk Units: 1 year
- Qty per Unit: 1200 (draining per year)
- Total Wk Qty: 1200 drainings
- Daily Wk Qty: 5 drainings a day

Example 5:

- Wk Units: 1200 miles
- Qty per Unit: 1 (hour per mile)
- Total Wk Qty: 1200 hrs
- Daily Wk Qty: 5 hrs a day

The above examples are more like planning strategy to meet your budgets or your man hours.

Adjusting Labor days calculation

The screenshot shows the MAINTSTAR Work Planning Setup interface. Callout A points to the 'GET DEFAULT' table, specifically the 'Labor Days' row. Callout B points to the 'Labor / Equipment / Material' table. Callout C points to the 'Daily Work Qty' field. Callout D points to the 'Work Units' field. Callout E points to the 'Qty per Unit' field.

Work Planning Setup

Full menu: Home (BSMITH), Work Order, Work Request, Asset, Asset Management, Inventory, Reporting, Work Planning, Work Planning Setup, Actual Entry, Work Plan, Reports, Tools, Purchase Orders, Invoicing, Fleet, Setup, Security.

Navigation: (A) Work Plan, (L) Activity, (K) Management Unit, (M) Percentage Templates, (S) Calendar, (N) Activity Programs, (O) Labor Trades, (Q) Equipment Types, (P) Material Classes, (R) Pay Types, (X) Funding Sources.

Open Search: + NEW, SAVE, +, +, DUPLICATE, DELETE

Work Plan No. 0000001189, 2021, 01/26/2021

Mgmt Unit: 00003 - Public Works

Activity code: 100 - OUTSIDE CONTRACTOR

Activity Program: null - null

Labor / Equipment / Material

ADD	Edit	No	Type	Code	Description	Qty	Rate	Cost	UOM
☐	Edit	1	Labor	AA	ADMINISTRATIVE ASSISTANT	2.00	44.44	88.88	HR
☐	Edit	2	Equip	Car	4 door sedan 123	10.00	20.00	200.00	HR
☐	Edit	3	Material	1/2AC	1/2 ASPHALT CONC	10.00	50.00	500.00	TON

Labor: \$88.88, Equipment: \$200.00, Material: \$500.00

GET DEFAULT

		Total %	01/01 01/31	02/01 02/28	03/01 03/31	04/01 04/30	05/01 05/31	06/01 06/30	07/01 07/31	08/01 08/31	09/01 09/30	10/01 10/31	11/01 11/30	12/01 12/31
Edit	Percent	99.96	8.33	8.33	8.33	8.33	8.33	8.33	8.33	8.33	8.33	8.33	8.33	8.33
	Labor Days	801.00	66.75	66.75	66.75	66.75	66.75	66.75	66.75	66.75	66.75	66.75	66.75	66.75
	Crew Days	400.50	33.38	33.38	33.38	33.38	33.38	33.38	33.38	33.38	33.38	33.38	33.38	33.38

Mgmt Unit Budgets: \$1,125,821.52

	Total %	01/01 01/31	02/01 02/28	03/01 03/31	04/01 04/30	05/01 05/31	06/01 06/30	07/01 07/31	08/01 08/31	09/01 09/30	10/01 10/31	11/01 11/30	12/01 12/31
Labor	800.64	66.72	66.72	66.72	66.72	66.72	66.72	66.72	66.72	66.72	66.72	66.72	
Employees	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	

Unit Cost: \$2,811.04

Time(Hr) per unit: 16.00

Crew Size: 2.00

Crew Days: 400.50

Lump Sum Budget: .00

Non General Fund: .00

Planned Budget: \$1,125,821.52

To change the labor days, you must adjust the Qty per Unit as follows:

Labor Days Desired (A)

$(A / B * C / D = E)$

Actual Work Entry

Planning menu, Option F.

The Actual work entry screen has two purposes:

- It is used to make new work planning entries for Labor, Equipment, Materials, and Work Quantity quickly. The entries are made in a grid like the LEM section of Work Orders.
- It is used to retrieve previous entries for editing

The screen is divided into the top half and the bottom half. The bottom half is a grid used to make new entries or to display and edit entries that were previously made. The top half of the screen is used to set up common values for several of the fields in the bottom half grid. By setting them up first in the top half, there are fewer fields that will need to be selected for the entries in the bottom half. This helps you make the entries quicker and easier.

Setup area

Work Planning entries

Edit	Work Track #	Type	Code/ID	Description	Pay	Hours	Rate	Amount	Date	Mgmt Unit	Activity	Location
☐	0001514540	Labor	R33	RANEY RAY	R	5.00	157.25	\$686.25	01/27/2020	00001221	N57	SRPAL29330
☐	0001514541	Labor	BA4	BALANAY MELCHOR	R	5.00	102.46	\$512.30	01/27/2020	00001221	N57	SRPAL29330
☐	0001514542	Labor	CA4	LOUIE CERVANTES	R	5.00	102.46	\$512.30	01/27/2020	00001221	N57	SRPAL29330
☐	0001514543	Labor	N06	GREGORY NUNEZ	R	5.00	102.46	\$512.30	01/27/2020	00001221	N57	SRPAL29330
☐	0001514544	Labor	V04	VERDILE ANTHONY	HHTD	5.00	105.45	\$527.25	01/27/2020	00001221	N57	SRPAL29330
☐	0001514545	Equipment	L81K	TRUCK WITH SERVICE BODY	CREWCB	5.00	16.94	\$84.70	01/27/2020	00001221	N57	SRPAL29330
☐	0001514546	Equipment	K81B	4X2 EXTENDED CAB PICKUP TRUCK	PU3/4	5.00	6.23	\$31.15	01/27/2020	00001221	N57	SRPAL29330
☐	0001514547	Equipment	K87C	2015 FORD F-250 XL 4X2 PICKUP	PU3/4	5.00	6.23	\$31.15	01/27/2020	00001221	N57	SRPAL29330
☐	0001514548	Equipment	P88H	2017 PETERBILT W/15 YARD DUMP	DMP10W	5.00	34.46	\$172.30	01/27/2020	00001221	N57	SRPAL29330
☐	0001514549	Equipment	Q81D	1993 ZIEMAN 2347A	TRAILERS	5.00	9.55	\$47.75	01/27/2020	00001221	N57	SRPAL29330
☐	0001514550	Equipment	R81B07	CASE800SM+ BACKHOE	BAKH0E	5.00	33.30	\$166.50	01/27/2020	00001221	N57	SRPAL29330
☐	0001514551	Work Qty	CUBIC YD			24.00	CUBIC YD	0.00	01/27/2020	00001221	N57	SRPAL29330
☐	0001514552	Labor	T42	TSURUI MATT	HFMS	5.00	140.81	\$704.05	01/27/2020	00001221	N57	SRGE10431C
☐	0001514553	Labor	R33	RANEY RAY	R	5.00	152.73	\$763.65	01/27/2020	00001221	N57	SRGE10431C
☐	0001514554	Labor	BA4	BALANAY MELCHOR	HPCL	5.00	137.25	\$686.25	01/27/2020	00001221	N57	SRGE10431C
☐	0001514555	Labor	CA4	LOUIE CERVANTES	R	5.00	102.46	\$512.30	01/27/2020	00001221	N57	SRGE10431C
☐	0001514556	Labor	N06	GREGORY NUNEZ	R	5.00	102.46	\$512.30	01/27/2020	00001221	N57	SRGE10431C
☐	0001514557	Equipment	L81K	TRUCK WITH SERVICE BODY	CREWCB	5.00	16.94	\$84.70	01/27/2020	00001221	N57	SRGE10431C
☐	0001514558	Equipment	K50G	2013 FORD F-250 4X2 SUPERCAB P	PU3/4	5.00	6.23	\$31.15	01/27/2020	00001221	N57	SRGE10431C
☐	0001514559	Equipment	K87C	2015 FORD F-250 XL 4X2 PICKUP	PU3/4	5.00	6.23	\$31.15	01/27/2020	00001221	N57	SRGE10431C

To make Actual Work Planning entries:
Make sure the bottom half of screen is clear. Use clear field button if needed.

- Set the common values for your entries in the fields at the top
- Set the date
- Set the Management Unit and Activity
- Set Location code and account number if used.
- Click add entry at the bottom area.

A new line will appear with the entry type set for labor because labor is the most common entry. Complete the line by choosing the technician's name in the code/ID field. Enter Hours worked (use fractions, 1.5 for an hour and a half). Pay type if different from the default. The values from the top half should be filled in on the bottom automatically.

To add a second line, you can use the 'add entry' button or press the enter key. The next line will appear, and the same type will be selected. In this example since we made a labor entry, the next line will be labor. When you are ready to change to another type of entry such as equipment or materials,

you can change the type drop down. The field will change by typing the first letter such as L for labor, E for equipment, M for materials etc.

Complete the fields for this entry such as the Id for the person and hours etc. You can also add crews with the add crew button. (Crews are created in the Work Order menu section. Follow the same pattern for equipment, materials and contractor.

Work Quantity Entry

Work Quantity entry is dependent on the activity chosen. Different activities have different units of measurements for Work Quantity. Because of this, make sure your entry before the work quantity, has the correct activity code listed. The Work Quantity will be displayed with the correct unit of measure for the previous activity. Example activity potholes, unit of measure might be a count of potholes. Striping might have a unit of measure the number of miles of striping performed etc. The Validate button, saving the entries.

The validate button can be used to check that your entries have entered correctly and required fields have been entered correctly. When you save the entries, all correctly entered entries will go away on the screen. They will be saved in the database. IF a line has a problem it will remain on the screen for you to correct and save again.

APPLY ALL Button: **USE WITH CAUTION!** When the top half window is filled-in with all the information, you can use this button to apply all the top information to all your entries on the bottom half screen. Again, it will **APPLY TO ALL OF YOUR ENTRIES, NOT ONE**. If you need to make one change to your entry, just go directly to your entry and make the change to simplify your confusion.

Actual Entry - Work Order Entry

PLEASE NOTE: THIS FUNCTIONALITY HAS CHANGED SIGNIFICANTLY IN VERSION 15

Actual entries can record activities that are part of a work order or are stand alone. Actual entries can be used to record events that DO NOT have a related work order. For example, If you want to record some activities that have no asset but you want to record time and hours, such attending meetings, or time off etc., the Actual entry screen is a great way to track these activities.

But if you DO have a work order already started, you can make your LEM entries here on the Actual entries screen if you prefer, instead of making the entries on the work order. To do this, the work order should be created, and you should know the work order number that you are recording time, materials, and work quantity to.

You will see a work order field at the top section and on the individual entry lines.

When you have the Work Order identified in your entry lines, (Labor, Equipment, Material, Contractor, and Work Quantity) THE ENTRIES WILL BE PUSHED TO THE WORK ORDER WHEN YOU SAVE.

This is a NEW function and very powerful. You now have the choice of entering your LEM entries on the work order directly OR using the very popular Actual entry screen.

Entry Tips

This menu choice is designed to enter actual work completed. The entry can be done daily, weekly etc. Tips for speedy data entry. If you would like to eliminate mouse usage for drop down windows data selection, for example in a Type field enter first character (example L for Labor or Alt + Arrow Down). The same for Code and other dropdown fields.

Use the enter key to indicate that a data entry is completed in a current entry line and to create a new line of entry. Some previously entered information will be defaulted from the previous line of entry, to speed up repeatable data entry.

To delete a record which is not currently displayed, click on Find to select the record then click on Scissors on a Tool Bar to delete. Then click on Save. If the top half screen and bottom half screen are too confusing for you, please ignore the top half screen and just make your entries right on the bottom half screen.

CLEAR FIELDS button:

is used to clear the top half window; so when the new button is clicked, there will be no default data filled-in in the fields of your new entry.

Retrieving Entries

The actual entry screen can be used to retrieve entries already made for purpose of review or editing. RETRIEVE button: retrieving old transactions for editing purposes. The Fields at the top of the screen can act as filters to search existing records. Fill in the fields related to your search such as:

- Dates from and to, for a certain date range.
- Select a particular management Unit or activity code.

If you leave any field empty, the empty field will not be filtered and all values for that field will be shown. First, do a lookup and pick an entry, then Retrieve will retrieve related entries that have matching Admin System and Activity. Click the Green Retrieve button to activate the search. The results will be shown in the table.

Set search values here

Retrieve the search results

Edit	Work Track #	Type	Code/ID	Description	PayType / Class	QTY	UOM	Unit Cost	Total Cost	Date	Mgmt Unit	Activity	Location	Sub
☐	0001941974	Labor	564		R	2.50	HR	0.00	\$0.00	08/20/2020	0000712	N57		
☐	0001942037	Labor	999	SUB CONTRACTOR	R	0.00	HR	0.00	\$0.00	01/23/2021	0000712	604		
☐	0001942038	Labor	880	NATALIE BAKER	R	0.00	HR	102.44	\$0.00	01/23/2021	0000712	G00001		
☐	0001942039	Labor	880	NATALIE BAKER	R	0.00	HR	102.44	\$0.00	01/23/2021	0000712	G00001		
☐	0001942040	Labor	BA4	MEL BALANAY	R	0.00	HR	137.25	\$0.00	01/23/2021	0000712	G00001		
☐	0001942042	Labor	880	NATALIE BAKER	R	0.00	HR	102.44	\$0.00	01/25/2021	0000712	602		
☐	0001942043	Labor	A15	ADAMS JACK	R	1.00	HR	137.25	\$137.25	01/25/2021	0000712	N24		
☐	0001942045	Labor	A21	ALDRIDGE LEONARD	R	2.00	HR	110.73	\$221.46	01/26/2021	0000712			
☐	0001942047	Labor	T42	MATTHEW TSURUI	R	2.00	HR	140.81	\$281.62	01/29/2021	0000712	N57		
☐	0001942048	Equipment	L53C02	2002 FORD F550 SUPER DUTY	CHPTK	2.00	HR	10.68	\$21.36	01/29/2021	0000712	N57		

In Editing Mode:

RETRIEVE button: retrieving original old transactions for editing purposes. First, do a lookup and pick an entry, then Retrieve will retrieve and filter related entries that have matching Admin System, Activity,

and Location. If different changes are made to these related transactions, retrieve feature will not consider them as original transactions, and will not retrieve them as related transactions.

CLEAR FIELDS button: to clear the information on the top half screen.

CLEAR SCREEN button: Clears the bottom half of the screen

Work Planning Reports

Rollup/Summary & Group Length Concept:

This feature is for the reports and it works the same throughout MaintStar wherever it is available.

- Rollup/Summary: is used when multiple Adm Systems are selected.
- Group Length: is how far you want to rollup

Example:

- Management Unit # 00001 is M&O maintenance
- Management Unit # 000012 is Road Administration
- Management Unit # 00001221 is Road maintenance

If we ran a report with these three management units selected and checked Rollup/Summary with Group Length of 5, then we will have the total cost maintenance for the park. As you can see M&O Maintenance #00001 has 5 digits. If we don't check Rollup/Summary, then we would see a detail report of all three Management Unit. In other words, Rollup/Summary provides the flexibility of summary cost report for any level. Let's look at another example below.

Management Unit / Resource		Total	07/01 07/31	08/01 08/31	09/01 09/30	10/01 10/31	11/01 11/30	12/01 12/31	01/01 01/31	02/01 02/28	03/01 03/31	04/01 04/30	05/01 05/31	06/01 06/30
00001 - M&O ADMINISTRATION														
SEDANS A - SEDANS A	Equip Days	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Available: 0.1	City Needs	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SUV - BLAZER EXPLORER DURANGO	Equip Days	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Available: 1.5	City Needs	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Available Labor for 00001	Equip Days	65.60	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Available	City Needs	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
00001221 - ROAD MAINTENANCE - LBS														
BAKHCE -	Equip Days	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Available: 1	City Needs	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BERM - BERM MACHINE	Equip Days	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Available: 1	City Needs	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BROOM - TRACTOR BROOM	Equip Days	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Available: 1	City Needs	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
CHIPPR - CHIPPER	Equip Days	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Available: 1	City Needs	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
CHIPTE - CHIPPER TRUCK	Equip Days	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Available: 1	City Needs	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Running Reports

Resource Requirements (Planning Menu Option B.)

MAINTSTAR Asset Management

WORK PLAN. REPORTS

(B) Resource Requirements (C) Work Planning Report (G) Actual Work Report (H) Work Status Report (J) Planne

☐ Rollup/Summary

Group Length 12 Year No 2021

Mgmt Unit

00001 x 000012 x 00001221 x

☐ Labor Requirements ☒ Equipment Requirements ☐ Material Requirements

This screen will give you an overview of the resource requirements such as Labors, Equipment, Materials, that are used under each Management Unit Number. There are three radio buttons for Labor Requirements, Equipment Requirements, Material Requirements.

- Management Unit: from the dropdown list select a management unit for the report. You can add additional management units. The ones you have selected appear below the field on the screen. Click on each Tab - Labor, Equipment, or Material, one by one to view your resource requirements report.
- Labor Requirements Radio Button
- This screen will show the Labor Resource Requirements for the Management Unit number you selected.
- The second column will show how many Labor Resources available for each type of labor. This available labor number was set up under Option J. (Management Unit - Labor Available tab).
- The third column is the header (label) for each row
- The fourth column shows the total number of workdays for each type of worker for your Management Unit # you selected. Ex: you could have Maintenance Worker for one category, and Supervisor for another. (Note: Workers and Supervisors are created under Option A – Labor Tab).
- (Yearly) Workdays = Number of Maintenance Worker * Crew days; the same for Supervisor, etc.
- The Total Workdays then will be divided by 12 to find Monthly Workday.
- Then take the number in each month divide by the number of Calendar Workdays (that was set up under Calendar Setup Option R., Current Year Button) to find the number of Employee for each day.
- Take the Resource Labor Available minus Employee each day to obtain the remaining available

- The Total Manpower: is the sum of all workdays and workers required for the plan. (see below)

MAINTSTAR All Modules
Labor Resource Requirements
Year: 2019

Detail / Summary	Management Unit / Resource	Total	07/01 07/31	08/01 08/31	09/01 09/30	10/01 10/31	11/01 11/30	12/01 12/31	01/01 01/31	02/01 02/28	03/01 03/31	04/01 04/30	05/01 05/31	06/01 06/30
00001 - M&O ADMINISTRATION														
AS2 - ADMIN SPECIALIST 2	Available: 1 Work Days Employees Available	216.04	18.99 0.00	18.04 0.00	18.04 0.00	19.01 0.00	16.01 0.00	18.99 0.00	16.96 0.00	16.03 0.00	19.01 0.00	18.99 0.00	16.96 0.00	19.01 0.00
DIR - DEPUTY DIRECTOR PIW	Available: 1 Work Days Employees Available	216.04	18.99 1.00	18.04 1.00	18.04 1.00	19.01 1.00	16.01 1.00	18.99 1.00	16.96 1.00	16.03 1.00	19.01 1.00	18.99 1.00	16.96 1.00	19.01 1.00
PCE - PRINCIPAL CIVIL ENGINEER	Available: 0.25 Work Days Employees Available	54.01	4.75 0.25	4.51 0.25	4.51 0.25	4.75 0.25	4.00 0.25	4.75 0.25	4.24 0.25	4.01 0.25	4.75 0.25	4.75 0.25	4.24 0.25	4.75 0.25
SE1 - SECRETARY I	Available: 1 Work Days Employees Available	216.04	18.99 0.00	18.04 0.00	18.04 0.00	19.01 0.00	16.01 0.00	18.99 0.00	16.96 0.00	16.03 0.00	19.01 0.00	18.99 0.00	16.96 0.00	19.01 0.00
TMP - TEMP CLERKS	Available: 2 Work Days Employees Available	432.00	38.02 2.00	35.94 2.00	35.94 2.00	38.02 2.00	31.96 2.00	38.02 2.00	34.04 2.00	31.96 2.00	38.02 2.00	38.02 2.00	34.04 2.00	38.02 2.00
Available Labor for 00001	Work Days 5.25 Employees Available	1,134.13	99.74 5.25	94.57 5.25	94.57 5.25	99.80 5.25	83.99 5.25	99.74 5.25	89.16 5.25	84.06 5.25	99.80 5.25	99.74 5.25	89.16 5.25	99.80 5.25
00002 - ROAD ADMINISTRATION - L&B														
CL - CREW LEADER	Available: 0 Work Days Employees Available	11.96	0.96 -0.07	1.04 -0.07	0.96 -0.07	1.04 -0.07	0.90 -0.07	1.04 -0.07	0.96 -0.07	0.90 -0.07	1.04 -0.07	1.04 -0.07	1.04 -0.07	1.04 -0.07
FMS - FIELD MAINTENANCE SUPERVISOR	Available: 0 Work Days Employees Available	5.11	0.40 0.03	0.45 0.03	0.40 0.03	0.45 0.03	0.38 0.03	0.45 0.03	0.40 0.03	0.38 0.03	0.45 0.03	0.45 0.03	0.45 0.03	0.45 0.03

Equipment Requirements Tab

This screen is almost the same as the Labor screen above.

- To obtain the Equipment Days: Take the Equipment Quantity times Crew Days. (Equipment Qty was created under Option A – Equipment Tab),
- Take the total Equipment Days divide by 12 to find the number of days the equipment is used in a month.
- Take the number in each month divide by the number of Calendar Working Days = Qty needed a day.

Material Requirement Tab

This screen shows the total number of Material Requirement for the whole year.

- Total = Material Qty * Crew Days (this was created under Option A., Material tab)
- Monthly Material = Total / 12

Work Planning Reports

Planning Menu, Option C.

Condition Tab

This screen will allow you to run reports by Multiple Management Unit, or by Multiple Activities, or by Multiple Programs. If you leave all boxes blank, the report will show all your Work Planning, and that will take a bit longer time to run if you have a big database.

Rollup/Summary and Group Length check box:

Group Length is part of Rollup/Summary. The Group Length option is unavailable when the Rollup box is not checked. If Rollup box is not checked, the report will show the data in detail by default.

MAINTSTAR Asset Management

WORK PLAN. REPORTS

(B) Resource Requirements (C) Work Planning Report (G) Actual Work Report (H) Work Status Report (I) Planned Use Summary (Ja) 3 Year Planned Use Summary

☐ Rollup/Summary

Group Length 12 Year No 2021 Mgmt Unit All

Activity All Program All Funding Source All

☒ Work Plan Report ☐ Percent/Labor Days Distribution

MAINTSTAR Asset Management

WORK PLAN. REPORTS

(B) Resource Requirements (C) Work Planning Report (G) Actual Work Report (H) Work Status Report (I) Planned Use Summary (Ja) 3 Year Planned Use Summary

☐ Rollup/Summary

Group Length 12 Year No 2021 Mgmt Unit All

Activity All Program All Funding Source All

☒ Work Plan Report ☐ Percent/Labor Days Distribution

By Mgmt Unit By Activity By Program By Funding Source

Work Plan by Mgmt Unit

Year: 2021

Management Unit / Program / Activity	Inv Qty	Inv Unit	Qty Per Unit	Total Work Qty	Work Unit	Avg Daily WkQty	Crew Size	Crew Days	Labor Days	Labor \$	Equipment \$	Material \$	Total	Factor
00007 - ACFWA M&O														
01 - GENERAL ADMINISTRATIVE														
601 - ADMINISTRATION	1.00	MGT UNIT	6,920.00	6,920.00	LBR HOURS	40.00	4.00	173.0	692.0	\$753,675	\$11,816	\$0	\$165,509	N
602 - CLERICAL	1.00	MGT UNIT	1,730.00	1,730.00	LBR HOURS	10.00	1.00	173.0	173.0	\$108,762	\$0	\$0	\$108,762	
								346.0	865.0	\$862,437	\$11,816	\$0	\$274,271	
000071 - ROADS ADMIN - SBS														
01 - GENERAL ADMINISTRATIVE														
604 - GENERAL OVERHEAD	1.00	MGT UNIT	1,730.00	1,730.00	LBR HOURS	10.00	1.00	173.0	173.0	\$224,312	\$11,816	\$0	\$236,129	N
								173.0	173.0	\$224,312	\$11,816	\$0	\$236,129	
0000712 - ROADS MAINTENANCE - L&B														
01 - GENERAL ADMINISTRATIVE														
604 - GENERAL OVERHEAD	1.00	MGT UNIT	1,730.00	1,730.00	LBR HOURS	10.00	1.00	173.0	173.0	\$184,349	\$0	\$0	\$184,349	N
02 - FLOOD CONTROL MAINTENANCE														
M02 - EROSION REPAIR	101.00	CH MILES	4.95	499.95	CUBIC YD	25.00	5.50	20.0	110.0	\$91,755	\$34,138	\$185,900	\$314,875	N
04 - CRITICAL RESPONSE														
N13 - STORM RESPONSE & CLEANUP	140.00	INSPECT	7.20	1,008.00	LBR HOURS	50.00	5.00	20.2	100.8	\$83,717	\$14,629	\$0	\$97,750	N
N45 - SLIDE REMOVAL	140.00	RD MILES	22.86	3,199.70	CUBIC YD	80.00	8.25	40.0	330.0	\$270,996	\$74,604	\$0	\$345,596	N
N52 - GUARD RAIL REPAIR	1.00	MGT UNIT	2,750.00	2,750.00	LN FT	25.00	5.00	110.0	550.0	\$421,867	\$33,726	\$0	\$455,593	N
05 - ENGINEERING/INSPECTION/PROJECT														
M02 - INQUIRY/INSPECTION	1.00	MGT UNIT	157.66	157.66	INQUIRY	3.00	1.00	52.6	52.6	\$49,300	\$0	\$0	\$49,300	N
09 - MISCELLANEOUS MAINTENANCE														
S61 - FLAGGING	1.00	YEAR	1.00	1.00	LBR HOUR	40.00	4.00	0.0	0.1	\$73	\$0	\$0	\$73	N
10 - ROW & DRAINAGE PRESERVATION														

- Labor = Rate * 8hrsaday * Crew Days
- Equipment = Rate * 8hrsaday * Crew Days
- Material = Rate * Crew Days

Percent / Labor Days Distribution Tab

Select the radio button labeled Percent/Labor Days Distribution.

This screen will show the percentage and labor days distribution for each month throughout the whole year. This was done under option R., M. and K. The labor days come from your Working Planning (A), then divided by 12 months

Detail / Summary		Total	07/01 07/31	08/01 08/31	09/01 09/30	10/01 10/31	11/01 11/30	12/01 12/31	01/01 01/31	02/01 02/28	03/01 03/31	04/01 04/30	05/01 05/31	06/01 06/30
00007 - ACPWA M&O														
01 - GENERAL ADMINISTRATIVE														
601 - ADMINISTRATION	Labor Days	692.00	55.36	55.36	62.28	55.36	55.36	62.28	55.36	55.36	62.28	55.36	55.36	62.28
601 - ADMINISTRATION	Percent	100.00	8.00	8.00	9.00	8.00	8.00	9.00	8.00	8.00	9.00	8.00	8.00	9.00
601 - ADMINISTRATION	Labor Days	692.00	55.36	55.36	62.28	55.36	55.36	62.28	55.36	55.36	62.28	55.36	55.36	62.28
601 - ADMINISTRATION	Percent	100.00	8.00	8.00	9.00	8.00	8.00	9.00	8.00	8.00	9.00	8.00	8.00	9.00
602 - CLERICAL	Labor Days	173.00	13.84	13.84	15.57	13.84	13.84	15.57	13.84	13.84	15.57	13.84	13.84	15.57
602 - CLERICAL	Percent	100.00	8.00	8.00	9.00	8.00	8.00	9.00	8.00	8.00	9.00	8.00	8.00	9.00
602 - CLERICAL	Labor Days	173.00	13.84	13.84	15.57	13.84	13.84	15.57	13.84	13.84	15.57	13.84	13.84	15.57
602 - CLERICAL	Percent	100.00	8.00	8.00	9.00	8.00	8.00	9.00	8.00	8.00	9.00	8.00	8.00	9.00
000071 - ROADS ADMIN - S85														
01 - GENERAL ADMINISTRATIVE														
604 - GENERAL OVERHEAD	Labor Days	173.00	13.84	13.84	15.57	13.84	13.84	15.57	13.84	13.84	15.57	13.84	13.84	15.57
604 - GENERAL OVERHEAD	Percent	100.00	8.00	8.00	9.00	8.00	8.00	9.00	8.00	8.00	9.00	8.00	8.00	9.00
604 - GENERAL OVERHEAD	Labor Days	173.00	13.84	13.84	15.57	13.84	13.84	15.57	13.84	13.84	15.57	13.84	13.84	15.57
604 - GENERAL OVERHEAD	Percent	100.00	8.00	8.00	9.00	8.00	8.00	9.00	8.00	8.00	9.00	8.00	8.00	9.00
0000712 - ROADS MAINTENANCE - L08														
01 - GENERAL ADMINISTRATIVE														
604 - GENERAL OVERHEAD	Labor Days	173.00	13.84	13.84	15.57	13.84	13.84	15.57	13.84	13.84	15.57	13.84	13.84	15.57
604 - GENERAL OVERHEAD	Percent	100.00	8.00	8.00	9.00	8.00	8.00	9.00	8.00	8.00	9.00	8.00	8.00	9.00
604 - GENERAL OVERHEAD	Labor Days	173.00	13.84	13.84	15.57	13.84	13.84	15.57	13.84	13.84	15.57	13.84	13.84	15.57
604 - GENERAL OVERHEAD	Percent	100.00	8.00	8.00	9.00	8.00	8.00	9.00	8.00	8.00	9.00	8.00	8.00	9.00
02 - FLOOD CONTROL MAINTENANCE														

Actual Work Report

Planning Menu, Option G.

This powerful report has many ways it can be configured. The fields at the top of the screen filter the report data by date, Management Unit activity etc. If you do not set a value in a field, all data for that value will be displayed.

The radio buttons select 4 different versions of the report. Set the conditions you want them click run to display the report.

- Actual Work in summary and in detail format
- Cost in detail and summary

MAINTSTAR Asset Management

WORK PLAN. REPORTS

(B) Resource Requirements (C) Work Planning Report (G) Actual Work Report (H) Work Status Report (J) Planned Use Summary (Ja) 3 Year Planned Use Summary

Date From: 07/03/2019 To: 02/03/2021

Mgmt Unit: All Activity: All Program: All Funding Source: All Location: All Pay Type: All

Account: All Work Order: All Workgroup: All Project: All LEM Type: All LEM Code: All

Report Header:

☐ Actual Work Detail ☐ Actual Work Summary ☒ Cost Detail ☒ Cost Summary

Cost Report Summary
From 07/03/2019 to 02/03/2021

Mgmt Unit / Activity	Date	Location	Workgroup	Labor	Equip	Mat	Ctr	Misc	WkQty	Cost \$
00001 - M&O ADMINISTRATION										
										01 - GENERAL ADMINISTRATIVE
997 - SEMINAR/SAFETY/TRAINING	07/11/2019	UNKNOWN -	Work Planning	6.00	3.00	0.00	0.00	0.00	0.00	614.19
997 - SEMINAR/SAFETY/TRAINING	10/03/2019	UNKNOWN -	Work Planning	4.00	2.00	0.00	0.00	0.00	0.00	628.40
997 - SEMINAR/SAFETY/TRAINING	10/09/2019	UNKNOWN -	Work Planning	2.00	0.00	0.00	0.00	0.00	0.00	310.32
997 - SEMINAR/SAFETY/TRAINING	11/06/2019	UNKNOWN -	Work Planning	16.00	0.00	0.00	0.00	0.00	0.00	2,497.44
997 - SEMINAR/SAFETY/TRAINING	12/11/2019	UNKNOWN -	Work Planning	2.00	0.00	0.00	0.00	0.00	0.00	319.78
997 - SEMINAR/SAFETY/TRAINING	12/18/2019	UNKNOWN -	Work Planning	6.00	0.00	0.00	0.00	0.00	0.00	959.34
997 - SEMINAR/SAFETY/TRAINING	02/26/2020	UNKNOWN -	Work Planning	2.00	0.00	0.00	0.00	0.00	0.00	319.78
997 - SEMINAR/SAFETY/TRAINING	02/27/2020	UNKNOWN -	Work Planning	8.00	0.00	0.00	0.00	0.00	0.00	1,279.12
997 - SEMINAR/SAFETY/TRAINING	03/04/2020	UNKNOWN -	Work Planning	1.50	0.00	0.00	0.00	0.00	0.00	239.83
997 - SEMINAR/SAFETY/TRAINING	03/11/2020	UNKNOWN -	Work Planning	2.00	0.00	0.00	0.00	0.00	0.00	319.78
				49.50	5.00	0.00	0.00	0.00	0.00	7,487.98
										04 - CRITICAL RESPONSE
978 - EMERGENCY ADMIN COVID-19	03/18/2020	UNKNOWN -	Work Planning	8.00	0.00	0.00	0.00	0.00	0.00	1,218.32
978 - EMERGENCY ADMIN	03/19/2020	UNKNOWN -	Work	8.00	0.00	0.00	0.00	0.00	0.00	1,218.32

Work Status Report

Planning Menu, Option H.

This report runs very complicated and sophisticated calculations. This report runs dramatically faster in version 15. This screen gives you the report of Period to Date and Year to Date of all the plans you created under option A. See Option A for further info. and could be from W/O's. Run your Bi-Weekly reports to find out what w/o's were done for the work planning. Throughout our manual we use 8Hrs a day of work.

MAINTSTAR Asset Management

WORK PLAN. REPORTS

(B) Resource Requirements (C) Work Planning Report (G) Actual Work Report (H) Work Status Report (J) Planned Use Summary (Ja) 3 Year Planned Use Summary

☐ Rollup/Summary

Group Length: 12 Year No: 2020

Period From: 07/01/2019 Period To: 07/31/2019

Mgmt Unit: All Activity: All

Program: All Funding source: All

☒ Work Status (PTD/YTD) ☐ Work Calendar ☐ Overview

Work Status Report

☒ By Mgmt Unit ☐ By Activity ☐ By Program ☐ By Funding Source

Work Status (PTD/YTD) Report format (radio button): please see Work Planning (Option A) for further details of where the numbers come from.

- Plan Days = Labor Days / 12
- Act Days = from Actual Work Entry or from W/O's Labor Hrs. (4hrs or more = 1 day, 12hrs or more= 2days), or MISC. Cost. The Labor Date, and Misc. Date is used here, not issued date or target date. The same for parts. It is suggested that to enter all costs, except labor and material cost, under Misc. Cost Tab so that MaintStar can generate the productivity report based on the work quantity, and hours.
- Plan Qty = Daily Work Qty * Crew Days (option A)
- Act Qty = from Actual Work Entry or from W/O's Misc. Cost
- Labor Days Annual Plan = Labor Days (option A)
- YTD Actual are all from the actual entry and, if any, from wo's labor cost and misc. cost
- Work Qty Annual Plan = Total Work Qty (option A)
- YTD Actual are all from the actual entry and, if any, from wo's labor cost and misc cost
- Total Cost Annual Plan = Planned Budget (option A)
- YTD Actual are all from the actual entry and, if any, from wo's labor cost and misc cost
- Hour@Unit Annual Plan = Unit Hours (option A)
- YTD Actual are all from the actual entry and, if any, from wo's labor cost and misc cost
- Cost@Unit Annual Plan = Unit Cost (option A)
- YTD Actual are all from the actual entry and, if any, from wo's labor cost and misc cost

MAINTSTAR Asset Management

WORK PLAN. REPORTS

(B) Resource Requirements (C) Work Planning Report (G) Actual Work Report (H) Work Status Report (I) Planned Use Summary (J) 3 Year Planned Use Summary

Group Length: 12 Year No: 2020 Period From: 1 - 07/01/2019 Period To: 1 - 07/31/2019 Mgmt Unit: All Activity: All

Program: All Funding source: All

☒ Work Status (PTD/YTD) ☐ Work Calendar ☐ Overview

☒ By Mgmt Unit ☐ By Activity ☐ By Program ☐ By Funding Source

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MaintStar All Modules

Work Status Report by Mgmt Unit

From 07/01/2019 to 07/31/2019

Detail / Summary	Mgmt Unit / Program / Activity	Wk Unit	This Period Plan Days	This Period Act Days	This Period Plan Qty	This Period Act Qty	Total Cost Period Plan	Total Cost Period Actual	Unit Time(hr) Period Plan	Unit Time(hr) Period Actual	Unit Cost Period Plan	Unit Cost Period Actual	Labor Days Annual Plan	Labor Days Y-T-D Actual	Work Qty Annual Plan	Work Qty Y-T-D Actual
11 - PUMP STATION MTCE/REPAIR																
M26 - STRUCTURAL/MECH REPAIR		REPAIRS	0.0	1.1	0.0	1.0	\$0.00	\$1,364.00	0.0	9.0	\$0.00	\$1,363.80	0.0	1.1	0.0	1.0
			0.0	1.1			\$0.00	\$1,364.00					0.0	1.1		
			0.0	1.1			\$0.00	\$1,364.00					0.0	1.1		
00001 - M&O ADMINISTRATION																
01 - GENERAL ADMINISTRATIVE																
601 - ADMINISTRATION		LBR HOURS	114.1	0.0	912.4	0.0	\$83,850.00	\$0.00	1.0	0.0	\$91.90	\$0.00	1,296.0	0.0	10,368.0	0.0
602 - CLERICAL		LBR HOURS	19.0	0.0	151.7	0.0	\$4,657.00	\$0.00	1.0	0.0	\$30.70	\$0.00	216.0	0.0	1,728.0	0.0
997 - SEMINAR/SAFETY/TRAINING		LBR HOURS	0.0	0.8	0.0	6.0	\$0.00	\$614.00	0.0	1.0	\$0.00	\$102.37	0.0	0.8	0.0	6.0
			133.1	0.8			\$88,507.00	\$614.00					1,512.0	0.8		
04 - CRITICAL RESPONSE																
N17 - MATERIAL SPILL		SPILL	0.0	1.4	0.0	3.0	\$0.00	\$1,162.00	0.0	3.7	\$0.00	\$387.35	0.0	1.4	0.0	3.0
			0.0	1.4			\$0.00	\$1,162.00					0.0	1.4		
05 - ENGINEERING/INSPECTION/PROJECT																
M02 - INQUIRY/INSPECTION		INQUIRY	0.0	36.8	0.0	109.0	\$0.00	\$30,760.00	0.0	2.7	\$0.00	\$282.20	0.0	36.8	0.0	109.0
			0.0	36.8			\$0.00	\$30,760.00					0.0	36.8		
16 - TIME OFF																
955 - SICK LEAVE SELF		LABOR	0.0	1.0	0.0	8.0	\$0.00	\$755.00	0.0	1.0	\$0.00	\$94.40	0.0	1.0	0.0	8.0

Work Calendar Report format (radio button)

- Labor days = crew size * crew days
- Labor days then will be used to divide into 12months to obtain monthly labor days.
- Actual Daily Work Qty:
 - Crew Size (for any particular day) = total # of hours of all crew members (A) / by largest # of hours of any crew member (B)
 - Equivalent crew day (C) = total hours of all crew members (A) / (crew size (D) * Standard workday hours (E))

The above numbers are computed for each day per activity and admin. System. They are then summed by each activity for all days of time in question. The average daily production = sum of all work quantity (F) / sum of all equivalent crew days (C). An example for a 10-hour workday:

- 09/01/01 – Jim works 8 hours and Jake works 4 hours then leaves. Sara joins Jim and works 2 hrs. They install 5 signs.
- 09/02/01 – Jim works 5 hours and Jake works 3 hours and Sara works 5 hours. They install 4 signs.
- Calculations:

09/01/01: crew size (D) = (8+4+2) / 8 = 1.75; Equiv. crew day (C) = (8+4+2) / (1.75 * 10hrs/day) = .8

09/02/01: crew size (D) = (5+3+5) / 5 = 2.6; Equiv. crew day (C) = (5+3+5) / (2.6 * 10hrs/day) = .5

Work Qty (F) = 5 + 4 = 9 signs

Sum of Equiv. days = .8 + .5 = 1.3

ADP (average daily productivity or work qty) = 9 / 1.3 = 6.9

Overview report format (Radio Button): this screen will give you an overview of the last two screens.

Mgmt Unit / Program / Activity	Wk Unit	This Period Plan Days	This Period Act Days	This Period Annual Plan	This Period Y-T-D Actual	Total Cost Annual Plan	Total Cost Y-T-D Actual
10 - PUMP STATION MISC/REPAIR							
M26 - STRUCTURAL/MECH REPAIR	REPAIRS	0.0	1.1	0.0	1.1	0.00	1,364.00
			1.1	0.0	1.1		1,364.00
			1.1	0.0	1.1		1,364.00
00001 - MAO ADMINISTRATION							
001 - GENERAL ADMINISTRATIVE							
001 - ADMINISTRATION	LBR HOURS	114.1	0.0	1,296.0	0.0	902,801.00	0.00
002 - CLERICAL	LBR HOURS	19.0	0.0	216.0	0.0	53,000.00	0.00

Planned Use Summary Planning Manu, Option J.

This screen will give you a summary report of what you have created under Work Planning Option A comparing with what you have used under Option F. Select your conditions then click on the Result tab to run the report.

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MaintStar All Modules
PLANNED USE SUMMARY BY MANAGEMENT UNIT
Management Unit: 00007 - ACPWA M&O
From: 07/01/2020 To: 07/31/2020

Activity: 601 - ADMINISTRATION

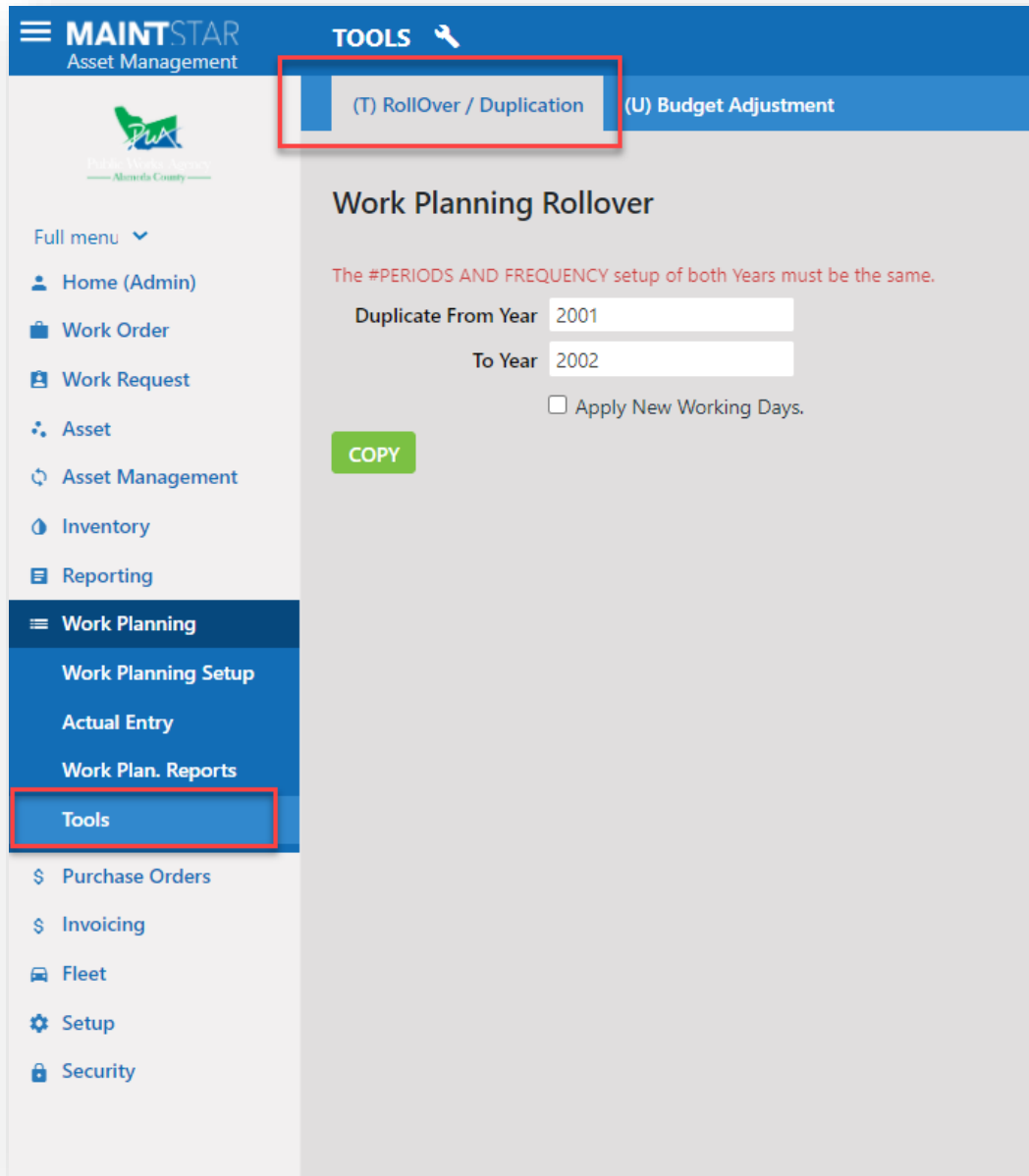
Planned		Actual	
LABOR	Qty		
DIR - DEPUTY DIRECTOR PW	1.00		
PT2 - PUBLIC WORKS TECH ASST 2	1.50		
PT3 - PUBLIC WORKS TECH ASST 3	1.00		
PWT - PUBLIC WORKS TECH ASST 1	1.00		
EQUIPMENT	Qty		
SEDANS A - SEDANS A	.08		
SUV - BLAZER EXPLORER DURANGO	1.50		

	ANNUAL PLAN	PERIOD PLAN	PERIOD ACTUAL
Inventory Qty	1.00		
Qty per Inv Unit	0.0000	612.0000	0.0000
Work Qty	7.650.00	612.00	0.00
DailyWork Qty	45.00	45.00	0.00
Crew Size	4.50	4.50	0.00
Crew Days	170.00	13.60	0.00
Unit Cost	\$2.36	\$2.36	\$0.00
Total Cost	\$18,046.18	\$1,443.69	\$0.00

Rollover Duplication Planning Menu, Option T

This screen allows you to duplicate your previous budget year to the new budget year. If "Apply New Working Days" box is checked, new working days for the new year will be applied automatically. If it is not checked, the new year will have the same working days as the old one.

BE CAREFUL!!! If you try to duplicate over an existing year, all you existing plans of that year will be deleted. You will be confirmed before deleting.



Budgets Adjustment Planning Menu, Option U.

This screen allows you to adjust the funds to fit your plans. It can be increased or decreased by the percentage or by the amount entered. You then can view the differences and decide to update the new rates or to cancel out without making any changes. We offer a quick backup function so you can back up before apply the new rates, and to restore right after at the moment you apply the new rates if you accidentally made a mistake. These backup and restore functions are not meant for database backup, and they will not work on different versions of MaintStar.

Retrieve button: select your Management Unit, then click on this button to obtain activity budgets.

- Final box: enter the final amount you want to see for all budgets to be calculated. The key here is to enter the increase/decrease percentage number or the final number, and either to calculate all budgets evenly or calculate only highlighted lines.
- Update W/Plan button: click on this button only if you are sure that you want to apply new working budgets and to overwrite the existing budgets.
- Reduce: if this box is checked, the planning cost will be reduced; if not checked, it will increase.
- Run Calculation: click on this button to view the new Total Working Budgets or the Final Budgets.
- Same Percentage: enter the percentage# to increase (box unchecked) / reduce (box checked) all activities equally (the original budgets) or highlight the specific activities to be calculated.
- Direct Input: enter the amount directly into the yellow box where you want to reduce or increase amount. The bottom Direct Input option and the top right Enter Data option have to sync.
- Priority/Multi%: goes with top direct input option. First enter your priority number in the Priority box, like 1 or 2 or 3, then setup Priority/Multi%, then run the calculation button.
- Highlight/Select Data allows you to run Calculations on the hilted lines only. Hold down Ctrl key and left click to highlight.
- Enter Data allows you to enter the data directly to the budget yellow box and finish it with a tab. Make sure the Top right Enter Data is selected.
- Apply New Rates: allows you to make changes (%) right on those small boxes for Labor, Equipment, Material rates setup, and click on Update DB button. Then to apply to the working planning, you must click on Update W/Plan button.

The screenshot displays the MAINTSTAR Asset Management interface, specifically the 'Tools' section under 'Budget Adjustment'. The interface includes a sidebar with navigation options like 'Home (Admin)', 'Work Order', 'Work Request', 'Asset', 'Asset Management', 'Inventory', 'Reporting', 'Work Planning', 'Work Planning Setup', 'Actual Entry', 'Work Plan, Reports', and 'Tools'. The main area shows a form for budget adjustment with fields for 'Mgmt Unit' (000071 - ROADS), 'Year No' (2021), and 'Highlight Activities'. A table lists activities with columns for 'No', 'Mgmt Unit', 'Activity', 'Original Budgets', 'Working Budgets', 'Enter Budgets', 'Priority', '%', 'Inventory', 'Qty@Unit', 'Work Qty', and 'Unit'. Below the table, there are input fields for 'Direct Input', 'Reduce' (checked), 'Final' (0.00), and 'Final Budgets - Original Budgets' (0.00). A red callout box points to the 'Final' input field with the text 'Enter Percentage or Final amount for Budget'.

No	Mgmt Unit	Activity	Original Budgets	Working Budgets	Enter Budgets	Priority	%	Inventory	Qty@Unit	Work Qty	Unit
1	000071	604	236129.08	236129.08	0.00		.00	1.00	MGT UNIT	1730.00	LBR HOURS
Total:			236129.08	236129.08	0.00						

- Select % Template allows to apply new % template without changing your template setup. Works with Update W/Plan button.
- Activity Distribution and Text Description: if needed, you must make changes in your Activities Distribution and text description first, then go here to apply the changes to the existing plans.
- Bottom Left Box: shows the different amount between the new working budget and original budget.
- Bottom Right Box: shows the total of the total and the bottom left box.